

Withdrawal notice

Algebraicity of Weil classes on split abelian eightfolds

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Withdrawn on October 6, 2026

The proof contains a sign error in the stabilization-trace argument used to obtain negative double points. In the manuscript's signed double-point convention, write $I(f_1) = -m$ with $m > 0$. The proof assigns each reverse stabilization trace sign +1 and therefore claims that inserting m such traces makes the signed count zero.

Accounting for the opposite source orientations of the two branches of the standard cusp gives sign -1 for each reverse trace in this convention. The resulting count is therefore

$$I_{\text{new}} = I(f_1) - m = -2m, \text{ which is nonzero.}$$

The Eliashberg-Murphy cancellation theorem invoked at this step requires zero signed double-point count, so its hypothesis is not met. The subsequent oriented-surgery and embedded-brane construction is therefore unsupported, and the paper does not establish its claimed algebraicity theorem.

This withdrawal concerns the proof; it does not assert that the mathematical statement is false.

Archived manuscript

[Pre-withdrawal PDF](#)

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